Empathy in Real vs. Synthetic Clinician Comments — Bayesian Mixed-Effects Models

0.1 Scope

This document is a **companion to `report.Rmd`**, not a replacement for it - it only re-does the *mixed-effects modeling* stage (originally `3b_mixed_effects_models.R`, fit with `ordinal::clmm` / `lme4::glmer`), using `brms`/Stan instead. Everything else - the data description, QC, and Mann-Whitney tests - lives in the main report; this document doesn't repeat it.

Why: on the full dataset, Emotional Reactions' Model B (the model x method interaction) fails to converge under maximum likelihood - the fit's Hessian is not positive definite, so no valid standard error or p-value exists for it at all (see `report.Rmd`'s convergence discussion). The usual frequentist fix for this kind of separation, Firth-style penalized likelihood, has no mature implementation that supports *both* an ordinal cumulative-link outcome and a random intercept for post-clustering, in R or Python (checked directly: `ordinal::clm/clmm` have no bias-reduction option; `brglm2` only covers fixed-effects GLMs with no cumulative-link family). A Bayesian fit with weakly-informative priors resolves the same separation problem the same way Firth's correction does - by keeping estimates finite instead of letting them run to a boundary - while keeping the interaction and the `post_id` random intercept intact.

Data and response weights. All 137,278 comments are modeled (8,625 real, 128,653 synthetic, from 6,600 threads): no length-outlier exclusion and no random subsampling (both were examined and rejected; see `report_outlier_exclusion`). Synthetic comments outnumber real ones within a thread, because they were generated against each thread's total comment count while the real comments are a smaller clinician-only subset. To stop long threads from counting for more on the synthetic side than on the real side, each real comment gets weight 1 and each synthetic comment gets `w = min(1, real count / synthetic count)` for its (thread, model, method) cell. The synthetic weights sum to 77,603 - exactly what per-thread subsampling would keep - and the expected subsampled log-likelihood equals the weighted one, so a weighted fit targets the average over random subsamples without depending on any single draw.

`brms` multiplies each comment's log-likelihood by its weight, so this is a pseudo-likelihood: point estimates target the same quantity as subsampling, but posterior widths are approximate (they behave as if the sum of the weights were the sample size) and are not corrected for the weighting. A check that a fit with weight 2 on every comment matches an unweighted fit on the doubled data confirmed the weights enter the likelihood as intended. In a separate frequentist check on Interpretations and Explorations (50 balanced subsamples vs. one weighted fit), the weighted estimates sat a median of 0.65 SE from the subsample average (up to 3.4 SE), and the weighted fit's naive standard errors were about 12% larger than a single subsample's. The weighted fit is therefore a different estimator, not an exact stand-in for subsampling, and that check did not cover Emotional Reactions. Subsampling sensitivity checks are planned separately.

Models, fit separately for each mechanism, with weights `w` as above:

- **A** (`label | weights(w) ~ group + log(length) + (1|post_id)`, all comments): each (model, method) group vs. real.
- **B** (`label | weights(w) ~ model * method + log(length) + (1|post_id)`, synthetic comments only): simple-effects contrasts between models within each method, and between methods within each model.

Emotional Reactions uses `family = cumulative("logit")` (all 3 empathy levels); Interpretations and Explorations use `family = bernoulli("logit")` on `label == 2` (both never predict label 1 in this data - same as the frequentist version).

Table 1: Convergence for each brms fit: max Rhat < 1.01, minimum bulk-ESS ratio above 0.1, and zero divergent transitions.

Mechanism	Model	Family	Converged	Max Rhat	Min ESS ratio	Divergences
Emotional Reactions	A_group_omnibus	cumulative	TRUE	1.0099	0.215	0
Emotional Reactions	B_model_x_method	cumulative	TRUE	1.0098	0.198	0
Interpretations	A_group_omnibus	bernoulli	TRUE	1.0026	0.228	0
Interpretations	B_model_x_method	bernoulli	TRUE	1.0085	0.223	0
Explorations	A_group_omnibus	bernoulli	TRUE	1.0031	0.204	0
Explorations	B_model_x_method	bernoulli	TRUE	1.0096	0.238	0

Table 2: Model A: posterior median log(length) coefficient (logit scale), with its credible interval and probability of direction.

Mechanism	log(length) coef.	95% CrI (low)	95% CrI (high)	pd
Emotional Reactions	0.555	0.489	0.620	1
Interpretations	1.329	1.255	1.402	1
Explorations	-0.224	-0.280	-0.169	1

Priors: `normal(0, 2.5)` on every fixed effect and (for Emotional Reactions) the ordinal thresholds - a standard weakly-informative default for logistic-type models (Gelman, Jakulin, Pittau & Su, 2008) - and `student_t(3, 0, 2.5)` on the `post_id` random-effect SD (brms's own default). This is what actually addresses separation: brms's *default* fixed-effect prior is an improper flat prior, which would leave the same non-identifiability problem the frequentist fit has.

Sampler and reproducibility: `brms`/Stan via `cmdstanr`, 4 chains of 2000 iterations (1000 warmup), with 4 threads per chain. The Interpretations and Explorations Model A fits were rerun with 4000 iterations (same warmup) because their max Rhat was 1.011 at 2000. The seed (20260101) is fixed, but within-chain threading makes exact reproduction of the draws from the seed alone unreliable (identical seeds with threading gave different draws in a direct test, because multithreaded summation order is not deterministic). The fitted models, including every posterior draw, are saved to disk (`models_bayesian/`), with the software versions and settings in `run_metadata.txt`; reproducibility rests on those saved fits, not on regenerating identical draws.

Reading this document vs. the frequentist one: there is no p-value in a Bayesian fit. In place of BH-adjusted p-values, contrasts below report a 95% highest-posterior-density (HPD) credible interval and `pd`, the posterior *probability of direction* - the share of posterior draws on the same side of zero as the point estimate. `pd` and a p-value answer different questions and aren't interchangeable, but as a rough calibration, `pd > 0.975` is comparable to two-sided `p < 0.05` (Makowski et al., 2019). A contrast is called **credible** below only when *both* `pd > 0.975` **and** the 95% CrI for the odds ratio excludes 1 - deliberately a bit more conservative than either alone, since `pd` can be extreme even for a tiny, practically negligible effect. Contrasts are odds ratios (link-scale estimates, exponentiated).

0.2 Convergence

All six fits (Model A and Model B for all three mechanisms) converged cleanly: max Rhat < 1.01, minimum bulk-ESS ratio above 0.1, and zero divergent transitions throughout. The largest max Rhat is 1.0099 (Emotional Reactions, Model A); 4 of 6 fits are above 1.005, so several sit close to the 1.01 cutoff rather than well inside it.

Emotional Reactions' Model B is the fit this document exists to try to rescue - **this is the fit the frequentist model couldn't produce at all** - it now has a full posterior, credible intervals, and simple-effects contrasts below, none of which existed in the frequentist report.

Table 3: Model A: each (model, method) group vs. real (odds ratio below 1 = lower odds of higher empathy than real; credible rows with OR above 1 shaded).

Mechanism	Group	Odds ratio	OR 95% CrI (low)	OR 95% CrI (high)	pd
Emotional Reactions	gemini_dg	0.310	0.272	0.348	1
Emotional Reactions	openai_dg	0.064	0.053	0.075	1
Emotional Reactions	grok_dg	0.027	0.020	0.035	1
Emotional Reactions	gemini_sag	0.390	0.348	0.437	1
Emotional Reactions	openai_sag	0.104	0.089	0.119	1
Emotional Reactions	grok_sag	0.051	0.041	0.063	1
Emotional Reactions	gemini_sagii	0.390	0.351	0.439	1
Emotional Reactions	openai_sagii	0.113	0.097	0.131	1
Emotional Reactions	grok_sagii	0.025	0.019	0.033	1
Interpretations	gemini_dg	0.457	0.391	0.526	1
Interpretations	openai_dg	0.635	0.570	0.708	1
Interpretations	grok_dg	1.807	1.595	2.042	1
Interpretations	gemini_sag	0.376	0.322	0.442	1
Interpretations	openai_sag	0.687	0.622	0.770	1
Interpretations	grok_sag	1.747	1.555	1.963	1
Interpretations	gemini_sagii	0.462	0.400	0.535	1
Interpretations	openai_sagii	0.634	0.572	0.708	1
Interpretations	grok_sagii	2.134	1.899	2.395	1
Explorations	gemini_dg	0.171	0.152	0.194	1
Explorations	openai_dg	0.037	0.030	0.046	1
Explorations	grok_dg	0.043	0.035	0.052	1
Explorations	gemini_sag	0.185	0.164	0.209	1
Explorations	openai_sag	0.021	0.016	0.028	1
Explorations	grok_sag	0.073	0.063	0.085	1
Explorations	gemini_sagii	0.164	0.144	0.185	1
Explorations	openai_sagii	0.031	0.024	0.039	1
Explorations	grok_sagii	0.037	0.030	0.045	1

0.3 Comment length

Comment length (log-transformed) is credibly nonzero for all 3 mechanisms (pd > 0.975, 95% CrI excludes 0), and not in the same direction: Emotional Reactions, Interpretations score higher with more words, while Explorations scores lower with more words.

0.4 Real vs. synthetic (Model A)

real scores credibly higher than every synthetic group for Emotional Reactions, Explorations; Interpretations is the exception—grok_dg, grok_sag, grok_sagii score credibly *higher* than real.

Refitting Interpretations' Model A without `log(length)` flips the posterior median direction for grok_dg, openai_sag, openai_sagii (3 groups), consistent with a length confound rather than a pure model/method effect for those groups.

0.5 Simple-effects contrasts (Model B)

Same rationale as the frequentist report: rather than pooled main-effect means (which would be misleading wherever the model x method interaction is credible), two sets of simple-effects contrasts - models compared within each method, and methods compared within each model.

Unlike the frequentist report - where Emotional Reactions was entirely absent from both simple-effects tables because its Model B fit was unusable - Emotional Reactions appears in both tables above.

Table 4: Model B: pairwise model contrasts within each method.

Mechanism	Method	Contrast	Odds ratio	OR 95% CrI (low)	OR 95% CrI (high)	pd
Emotional Reactions	dg	gemini - openai	6.598	5.328	8.285	1.000
Emotional Reactions	dg	gemini - grok	12.084	8.962	16.142	1.000
Emotional Reactions	dg	openai - grok	1.836	1.300	2.579	1.000
Emotional Reactions	sag	gemini - openai	5.587	4.400	6.954	1.000
Emotional Reactions	sag	gemini - grok	8.312	6.760	10.402	1.000
Emotional Reactions	sag	openai - grok	1.489	1.113	1.964	0.997
Emotional Reactions	sagii	gemini - openai	5.026	4.074	6.291	1.000
Emotional Reactions	sagii	gemini - grok	16.995	12.535	22.637	1.000
Emotional Reactions	sagii	openai - grok	3.371	2.390	4.769	1.000
Interpretations	dg	gemini - openai	1.173	0.999	1.370	0.978
Interpretations	dg	gemini - grok	0.228	0.199	0.263	1.000
Interpretations	dg	openai - grok	0.194	0.168	0.223	1.000
Interpretations	sag	gemini - openai	0.985	0.831	1.179	0.562
Interpretations	sag	gemini - grok	0.208	0.179	0.240	1.000
Interpretations	sag	openai - grok	0.211	0.182	0.242	1.000
Interpretations	sagii	gemini - openai	1.276	1.071	1.505	0.997
Interpretations	sagii	gemini - grok	0.194	0.169	0.222	1.000
Interpretations	sagii	openai - grok	0.151	0.131	0.177	1.000
Explorations	dg	gemini - openai	2.157	1.662	2.732	1.000
Explorations	dg	gemini - grok	5.942	4.747	7.264	1.000
Explorations	dg	openai - grok	2.759	2.036	3.711	1.000
Explorations	sag	gemini - openai	2.267	1.619	3.134	1.000
Explorations	sag	gemini - grok	2.855	2.383	3.397	1.000
Explorations	sag	openai - grok	1.264	0.880	1.801	0.892
Explorations	sagii	gemini - openai	1.413	1.060	1.985	0.986
Explorations	sagii	gemini - grok	6.020	4.854	7.557	1.000
Explorations	sagii	openai - grok	4.280	2.891	5.935	1.000

Table 5: Model B: pairwise method contrasts within each model.

Mechanism	Model	Contrast	Odds ratio	OR 95% CrI (low)	OR 95% CrI (high)	pd
Emotional Reactions	gemini	dg - sag	0.780	0.683	0.890	1.000
Emotional Reactions	gemini	dg - sagii	0.801	0.700	0.918	1.000
Emotional Reactions	gemini	sag - sagii	1.024	0.905	1.169	0.648
Emotional Reactions	openai	dg - sag	0.665	0.538	0.811	1.000
Emotional Reactions	openai	dg - sagii	0.609	0.496	0.745	1.000
Emotional Reactions	openai	sag - sagii	0.922	0.786	1.097	0.829
Emotional Reactions	grok	dg - sag	0.539	0.378	0.745	1.000
Emotional Reactions	grok	dg - sagii	1.124	0.760	1.637	0.719
Emotional Reactions	grok	sag - sagii	2.088	1.487	2.979	1.000
Interpretations	gemini	dg - sag	1.189	1.011	1.426	0.970
Interpretations	gemini	dg - sagii	1.008	0.865	1.192	0.531
Interpretations	gemini	sag - sagii	0.849	0.709	1.010	0.962
Interpretations	openai	dg - sag	0.997	0.902	1.102	0.524
Interpretations	openai	dg - sagii	1.097	0.989	1.219	0.960
Interpretations	openai	sag - sagii	1.100	1.001	1.209	0.975
Interpretations	grok	dg - sag	1.082	0.967	1.201	0.929
Interpretations	grok	dg - sagii	0.857	0.769	0.947	0.998
Interpretations	grok	sag - sagii	0.791	0.718	0.879	1.000
Explorations	gemini	dg - sag	0.960	0.835	1.120	0.697
Explorations	gemini	dg - sagii	0.997	0.855	1.161	0.515
Explorations	gemini	sag - sagii	1.038	0.894	1.215	0.683
Explorations	openai	dg - sag	1.009	0.709	1.447	0.520
Explorations	openai	dg - sagii	0.655	0.478	0.910	0.994
Explorations	openai	sag - sagii	0.648	0.448	0.918	0.991
Explorations	grok	dg - sag	0.461	0.368	0.583	1.000
Explorations	grok	dg - sagii	1.012	0.774	1.295	0.535
Explorations	grok	sag - sagii	2.197	1.726	2.800	1.000